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Intelligent process monitoring for flexible production

Integration of Physics-Informed Machine Learning and Active Learning

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In highly customized production environments with a high degree of product variety, traditional approaches to process monitoring are increasingly reaching their limits. These approaches are mostly based on static data sets or

Recurring process patterns can lead to inaccurate predictions and increased false alarms in dynamic production environments. This paper presents a concept for making discrete production process monitoring more flexible, based on the combination of Physics-Informed Machine Learning (PIML) and Active Learning (AL). In agile production environments, this approach not only enables the detection of anomalies but also allows the monitoring model to be automatically updated in the event of false alarms. This ensures that the monitoring system remains precise even under variable production conditions, reducing false alarms and thus contributing to improved Overall Equipment Effectiveness (OEE).

New challenges due to skills shortages and agile production environments

Manufacturing companies are facing
major challenges: skills shortage, rising wage and production costs, the trend towards individualization, and increasing international cost-

tendruck. So that companies in this
To remain competitive in the environment, new approaches and solutions must be introduced into production. Cost-effective production with high quality and flexibility is the key to sustainable value creation and thus to securing long-term
competitiveness in the global market [1].

Driven by the shortage of skilled workers [2], however, there is an expansion of responsibilities, so that the number of machines to be monitored also increases.
It increases. If a critical value is exceeded-
However, with each step, the efficiency of
Humans and machines [3], which in the long term leads to a loss of quality and monetary losses
This can lead to problems in production. Ins-
Especially during periods of high capacity utilization, not all processes can be carried out continuously.
They need to be monitored, making process monitoring systems indispensable.

Due to the trend towards product individualization, the proportion of single-unit and small-batch production is increasing [1]. In this environment, process data are not available to a sufficient extent for statistical monitoring. Consequently, new, more flexible approaches are needed. Modern process-

Monitoring systems must be able to accurately map complex relationships. They must adapt flexibly to different, changing conditions.

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Adapt product ranges and their individual process conditions while working as data-efficiently as possible.

This article evaluates existing approaches from industry and academia. Building upon

A concept for a monitoring system is presented using the milling process as an example, which integrates Physics-Informed Machine Learning (PIML) and Active Learning (AL) to enable continuous adaptation and optimization to changing production conditions.

Classification and evaluation of industrial solutions for process monitoring

To assess the industry situation, three of the highest-grossing German companies were surveyed.

German machine tool manufacturers in

The following companies were considered in the field of machining with regard to their advertised solutions: DMG MORI AG, Maschinenfabrik Berthold Hermle AG, and the

Chiron Group SE [4]. The evaluation and classification of the individual technologies for process monitoring are based on the following criteria:

ÿ *Temporal resolution*

The system must be able to handle processes with a high time frame.

To monitor the resolution level, it must predict a large number of points in time per unit of time to ensure detailed and high-frequency monitoring.

ÿ *Robustness*

The system must be robust against incomplete, inaccurate, or faulty data and also function under

These conditions ensure a high level of prediction accuracy.

ÿ *Adaptability*

The system must be able to adapt efficiently to process changes, for example, a changing product range, in order to function even under altered production conditions.

to ensure reliable monitoring.

ÿ *Data efficiency*

The monitoring model must use the available data efficiently and be able to extract as much information as possible from the existing data.

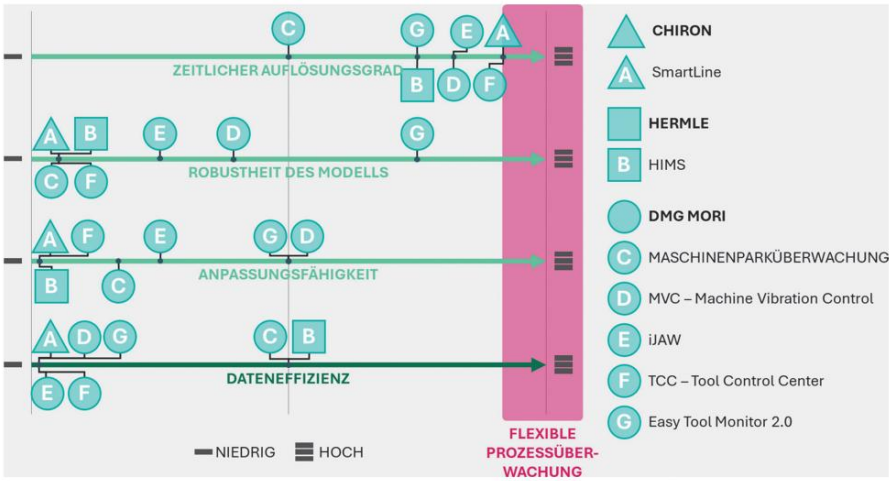


Figure 1. Qualitative classification of industrial solutions regarding intelligent process monitoring for the flexible production of the future.

Extracting data sets to achieve optimal results.

The overview shown in Figure 1 provides information on all technologies for which comprehensive information is available. The classification is based on defined criteria on a qualitative basis and takes into account the varying levels of information available. A low level of fulfillment corresponds to rigid process monitoring that depicts simple and recurring relationships.

A high degree of fulfillment represents the vision of flexible process monitoring.

Chiron offers a complete solution called *Smart-Line* (A), which consists of various digital components. The following *Smart-* Line modules are intended, among other things, to...

to help monitor processes [5].

The *ProcessLine module* simulates the real machining process using a digital twin that maps the geometry, kinematics and dynamics of an entire machining center.

This allows, for example, setup times to be optimized without blocking the machine. Building on this, the

The *ProtectLine* module reliably protects against production downtime by comparing the real and simulated workpiece geometry during the processes. Since the

If the digital twin collides first in the event of a machine crash, the machine can be brought to a safe standstill.

This is intended to provide a preventive and reliable to ensure protection against production outages. To ensure continuous processes.

To optimize processes, *Data-Line* captures and analyzes machine data in parallel with the process. This integrated machine and process diagnostics provides a basis for the continuous improvement of the entire process with regard to hardware and software, automation, and process and setup times. With the *ConditionLine* module...

The aim is to identify irregularities early through the permanent acquisition and analysis of machine data, so that maintenance and repairs can be planned and carried out proactively and tailored to the company's own production in order to avoid time-consuming and costly downtime [5].

Overall, the *SmartLine* (A) process monitoring modules exhibit low data efficiency. Specific data must be provided for each process.

The robustness of the model and the adaptability of individual models can be considered rather low. Through process simulation-

With the digital twin, a solution with a high-frequency resolution is already available.

Hermle presents the central monitoring tool *HIMS* (B) [6], which records and analyzes machine data to provide an overview of the live status of individual machines. As soon as anomalies or errors are detected, the responsible employee is notified.

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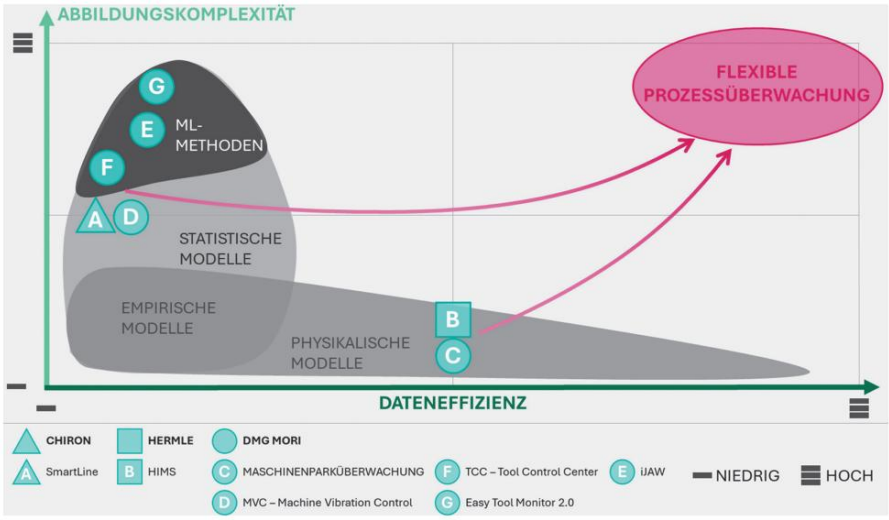


Figure 2. From rigid models to the vision of intelligent and flexible process monitoring: Classification of the considered solutions with regard to mapping complexity, which is composed of the temporal resolution, robustness and adaptability, and data efficiency.

Overall, the monitoring model is not very adaptable or robust, as it considers only a small number of data points and parameters. However, these can be monitored with a high temporal resolution. Since the model is based on a single machine and its...

Because it is tailored to specific processes, the data can be used efficiently compared to other technologies presented, so that the live status of the machines can be monitored with just a few data points.

In contrast to *Chiron* and *Hermle* *DMG Mori* presents a broad spectrum of technologies for process monitoring, which can also be used independently of one another. Consequently, this paper provides a separate evaluation and classification of these technologies. Similar to the previously presented technologies for machine park monitoring (C), *DMG Mori* also offers a runtime monitor [7, 8] to monitor machining cycles and thus optimize cycle times, as well as digital machine data acquisition [9], which enables clear performance monitoring of the entire machine park. These machine park monitoring technologies are tailored to specific processes and machines, so that only limited adaptability to changing conditions is possible.

Conditions and low robustness regarding changed data are present.

DMG Mori offers various Techno- logics for machine and process monitoring. With *MPC 2.0 – Machine Protection Control* [10, 11], vibrations and torques are measured during the process.

mente is monitored to enable a rapid shutdown in the event of a crash and to avoid damage.

Also *MVC – Machine Vibration Control* (D) [12, 13] detects vibrations in the machine using vibration sensors in the milling spindle. If the vibrations-

If vibrations become too strong, machining suggestions are made to reduce or eliminate them. An editing function allows for the adjustment of machining states, enabling continuous monitoring of process reliability. This allows for adaptation to slightly changed process conditions.

To monitor the tools and their clamping force, *DMG Mori* offers the following options: *iJAW* (E) [14, 15] monitors the clamping force of the tools during the process using sensors in a power chuck to ensure a controlled clamping force during machining. The sensors also allow for the monitoring of slightly altered process conditions and the processing of minor data deviations.

The *TCC – Tool Control* will be offered.

Center (F) [16, 17] provides chip detection for flat and tapered cutting surfaces and monitors the draw-in force. Cutting fractures can be detected by monitoring the symmetry of the bending moment. In addition

Radial and axial spindle loads can also be visualized in relation to the tool pull-in force to protect the machine and tool and to optimize tool usage. A high temporal resolution is achieved, but concrete data is required for accurate predictions, and no meaningful predictions can be made for changing process conditions. In contrast to the previously mentioned tool monitoring technologies, the *Easy Tool Monitor 2.0* (G) [18, 19] operates sensorlessly using automated learning from

Load limits for each tool used in the program. This prevents damage in the event of tool breakage or overload. Through automated learning-

In contrast to other technologies, nen offers high robustness and good adaptability, while at the same time offering low data efficiency.

The processes involved in machining are highly complex, making their monitoring a particular challenge. Suitable approaches adapted to the constantly changing product range are currently limited. The classification of the evaluated monitoring technologies shown in Figure 2 is based on the criteria of data efficiency and...

Mapping complexity, which comprises the temporal resolution, robustness, and adaptability. The range of solutions is characterized by two key features: high mapping complexity or high data efficiency. This leads to companies tending to offer rigid process monitoring systems tailored to a specific process.

In a flexible production environment However, with constantly changing product portfolios, such monitoring systems can reach their limits. Models must be continuously updated with less data.

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be adapted to new circumstances. For efficient process monitoring-

Intelligent models with high representational complexity and simultaneously high data efficiency are needed. Only in this way can monitoring systems adapt quickly to production conditions.

To identify more efficient solutions, current research approaches focusing on greater flexibility are being used.

and adaptability is examined.

Research approaches to increasing the flexibility of process monitoring systems

Due to the increasing number of product variants and the resulting demands-

Due to the demands on production flexibility, the requirements for traditional process monitoring systems have fundamentally changed in the context of Industry 4.0 [20–22]. Classical systems are based on empirically determined limit values and are typically designed to be reactive. While they can quickly detect serious errors such as tool breakage or machine failures and thus prevent further damage, they are unable to identify process-relevant errors early on and warn the machine operator in a timely manner. However, the early detection of errors with a short reaction time is crucial for quality-oriented and scrap-minimized production.

Induction. Especially with components containing

Given long production times, the timely detection of errors can significantly increase efficiency. Consequently, statistically based methods have already been developed.

investigated to identify potential anomalies early on [23]. These methods have already proven effective in mass production with recurring process patterns.

With increasing

The availability of process data allows machine operators to define intervention limits more precisely. Additionally, various sensors enable the detection of different types of anomalies. The trend shows that research has focused on specific problems. For example, acoustic and vibration sensors can be used to...

onssignals surface anomalies [24] or tool wear [25, 26] during machining can be identified-

the. Also the course of process forces

enables an assessment of the tool condition [27].

Correlations with the process forces could be established based on the spindle current signals.

can be observed [28, 29]. The particularly easy accessibility of spindle current signals enables cost-effective implementation in monitoring systems. Changes in these current signals over longer periods can also indicate tool wear [28]. Even short-term changes in the current signals allow conclusions to be drawn about process quality [30]. The introduction of machine learning methods enables the monitoring of even more complex manufacturing processes [31].

For example, recurring process steps can be identified using ML-based clustering, and their specific flow profile can be used as a reference pattern for

the monitoring of new products [32, 33].

Both statistical and model-based

However, these methods share a common weakness: the quality of the predictions depends

critically on the training data used [34]. In particular, due to the increasing individualization of

Dualization presents a major challenge in the production environment due to the sheer volume of data. When switching to new product variants, the validity of a model can no longer be guaranteed. In practice, this leads to false alarms, which in the worst case can result in machine downtime.

Approach to securing model validity in flexible Production environments

To improve forecast quality with fluctuating production volumes and high variant variation

To ensure diversity, process monitoring systems must become more flexible.

A model-based system is particularly suitable for dynamic adaptation to the current manufacturing process. However, the performance of the monitoring model, which provides the comparative value, remains limited by the lack of available field data for training. The production of a new workpiece directly impacts the validity of an initially trained model.

Dells out. The current process avoids the

When Generation Gn deviates from the original context represented in the training dataset due to changed boundary conditions, a “domain shifting” occurs [35]. However, a classically trained model can only make reliable predictions in already known domains [34]. To effectively solve this problem, domain-incremental learning can be used [35]. An effective approach for adapting the model to the new conditions is provided by PIML [36]. In addition, AL with integrated data augmentation can help to further improve the model quality [37]. The interplay of the system components is illustrated in Figure 3.

Subsystem 1 – Monitoring Model

The flexibility of a monitoring system begins with the establishment of reference values.

It should be as flexible as possible with regard to toolpath and machining parameters and be able to generate reference values with any desired temporal resolution.

This can be achieved through model-based monitoring. For example, a model [38] for predicting axis-specific current signals can be derived for monitoring CNC milling machines. By coupling a physical system description and a machine learning-based representation of the current value, the data efficiency of classical approaches can be combined with the mapping complexity of data-based approaches.

From input data read from the machine control, this hybrid model can be used to generate real-time setpoint current values.

The input variables used are acceleration, velocity, force, and the MRR (Material Removal Rate). Depending on sensor availability, the forces can optionally be calculated using a soft sensor.

This allows the system to be used for brownfield machines as well. The MRR data is determined through an integrated voxel-based material removal simulation. By integrating anomaly detection, the measured shaft currents can be compared with the calculated values. Process anomalies are used to increase the

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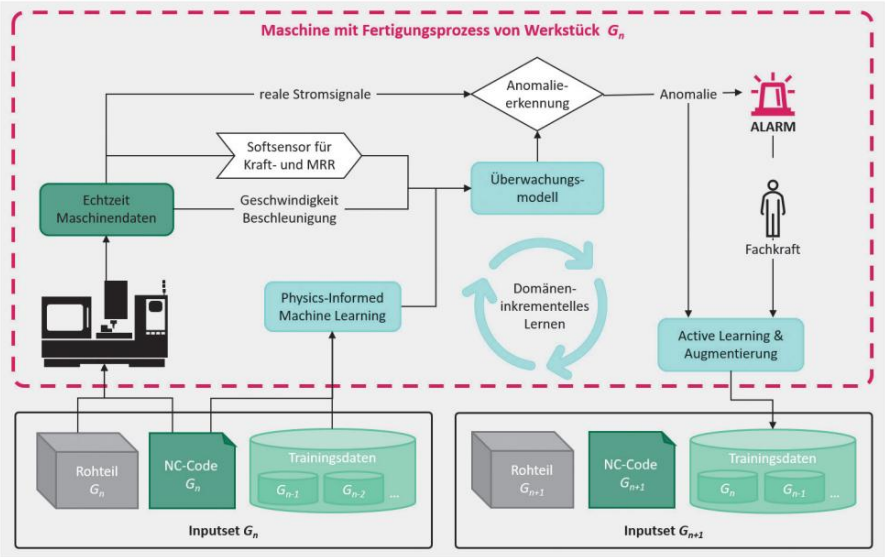


Figure 3. Concept of the model-based process monitoring system for flexible manufacturing. It consists of the subsystems monitoring model, PIML, and AL with integrated data augmentation, which interact across products through domain-incremental learning.

Robustness is summarized in intervals, which can be displayed and saved once a relevant size is reached. To identify these anomalies-

Depending on the requirements, a different strategy can be chosen for each interval.

Subsystem 2 – PIML

Due to the hybrid model structure [38], in addition to the overall model, the description of the physical equations is also provided. Although this is limited in its mapping complexity compared to a machine learning-based mapping, it is nevertheless able to extrapolate in a physically plausible manner due to the physical relationships.

Consequently, this can be used in a new process to generate support points until real post-production data is available. These enable improved extrapolation and can thus enhance the predictive accuracy of the overall model in previously unknown areas.

Significantly improve processes.

Before the manufacturing process begins, the machine's operating conditions are analyzed and missing process areas in the existing training dataset are identified. Based on the NC code, the input parameters are generated and the current values are determined using the...

A physical equation is generated. To complete the missing areas in the feature space, the generated data points supplement the existing training dataset. This targeted adaptation of the training dataset represents an integrated and continuous adaptation loop in the manufacturing process. Through this circular strategy, the process monitoring model is always adapted for the current workpiece generation Gn , thus effectively addressing a reduced data volume during product changes.

Subsystem 3 – AL with integrated data augmentation

If deviations occur between the actual and the predicted signal during the process, these are considered purely as process deviations in classical process monitoring.

However, particularly in a complex production environment, it must be assumed that anomalies are also attributable to poor prediction accuracy. Here, AI approaches can help to improve the monitoring model and avoid false alarms in the future. For this purpose, the intervals identified during the production process are analyzed. If an alarm occurs,

A specialist assesses whether the interval is a genuine one.

This is a process error. If this is not the case, the deviation was caused by data-based model errors and it

The interval data is annotated. This information is now available for model training.

From the perspective of the monitoring system are in particular data-based model-

Errors are undesirable because they can trigger false alarms. AL offers the possibility to specifically expand the training dataset in such areas.

To improve the model's accuracy in these areas, they are included in the training dataset for the next workpiece. Adding similar but not identical data points can lead to a further increase in robustness and generalizability [37, 39]. Data augmentation thus offers a way to further improve the model's predictive accuracy.

Thus, this subsystem offers an approach to continuously improve and further develop the model, even in cases of poor predictive accuracy, secured by the opinion of an expert.

This allows the monitoring system to continuously adapt to the actual production levels.

adapt to specific conditions without the need to specifically collect data through prior manufacturing.

Domain-specific incremental learning

The continuous product generation change leads to new process states, which means a change in the original training domain for the trained model [35]. Domain-incremental learning takes this change into account. The integration of the three subsystems results in the overall system shown in Figure 3. After each production process, an updated and balanced

A training dataset is available. Training always takes place before the production of the new generation, using the newly compiled and optimized training dataset. During the production process of a workpiece of generation Gn , the data of all previous generations are available.

Through continuous product generation-

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More and more ion changes are occurring
Data that is added to the training dataset as needed. The physics-based data augmentation uses the NC code of the current component as...

This forms the basis for supplementing missing areas in the feature space with synthetic data points. Based on this extended dataset, the monitoring model for the production of new components is trained and optimized. During the production process, intervals with anomalies are identified and issued as warnings. After the production process, specialists have the opportunity to analyze the identified intervals in detail. The data from intervals that are not attributable to process errors are augmented to ensure the highest possible robustness and generalizability.

In an ongoing production process with multiple generation changes, the system architecture enables a continuous training cycle, allowing for the fastest and most data-efficient adaptation of the monitoring system to new conditions. The training datasets are adapted and optimized for each new component before the manufacturing process. Additionally, the training datasets are expanded by AI after the manufacturing process to continuously improve the model's quality.

Discussion and Conclusion

Several process monitoring solutions are already available, but these are often rigid and limited to individual processes.
or process steps are tailored.
Such systems have already proven their worth in mass production and with product ranges featuring recurring machining patterns. Companies like *DMG Mori, Hermle ,* and *Chiron,* for example, already offer monitoring systems for individual process parameters. However, these systems are currently only partially able to adapt flexibly and efficiently to changing production conditions.

adapting to requirements. The trend towards increasingly complex and flexible production processes with ever shorter product lifecycles requires ever-

However, dynamic and adaptive monitoring models.

To increase the flexibility of monitoring models, the presented approach combines knowledge through two types of combined...
The training process integrates established strategies (PIML and AL). This allows for classification within an informed machine learning taxonomy [40]. The PIML strategy is based on knowledge and physical laws in the form of system equations. The AL strategy additionally incorporates expert knowledge via a human-machine interface.

The training system is integrated. This makes it possible to flexibly and data-efficiently develop monitoring models in an agile production environment, even without large amounts of production data.

operate.

However, several important aspects must be considered when applying this approach: The underlying model can only provide accurate predictions if high-quality training data is available. This depends heavily on the established system equation and the quality of the expert knowledge. Furthermore, it must be defined when a deviation should be considered an anomaly. Excessive sensitivity can lead to a large number of intervals being evaluated.

must. Therefore, there is a difference between a high To weigh sensitivity against the associated safety and the effort required for assessment.

Overall, the presented approach to intelligent process monitoring, combining hybrid monitoring models, PIML, and AL, offers a promising solution for process monitoring in flexible production environments. Compared to existing solutions, the presented approach considers both expert knowledge and physical laws. Continuous adaptation of the monitoring models to changing conditions allows process monitoring systems to be made more flexible, thereby reducing false alarms and increasing prediction accuracy. Future work will focus on investigating the individual subcomponents and their impact on flexibility.

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To optimize the performance of machine learning models.

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Abstract

Intelligent Process Monitoring for Flexible Manufacturing – Integrating Physics-In-formed Machine Learning and Active Learning-ing. As products and their production become more personalized and variant rich, traditional approaches to process monitoring are reaching their limits. These are typically based on static data sets or recurring process patterns, which can lead to inaccurate predictions and increased false alarms in agile production environments.

This article presents a concept for flexibilizing the process monitoring of discrete production processes based on the combination of Physics- informed Machine Learning (PIML) and Active Learning (AL). In agile production environments, this allows not only the detection of anomalies, but also the automatic update of the monitoring model in case of false alarms. As a result, the monitoring system remains accurate under variable production conditions, reducing the number of false alarms and contributing to im-proved Overall Equipment Effectiveness (OEE).

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Keywords

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Bibliography

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